



Reportedly the Linux Kernel is switching To Linux v4.0, and coming with many new addons: <http://dgit.in/1Do2V8G>

# Constraint Programming

>>When you think of constraints, the general feeling behind them is negative. Hopefully after reading this, you will think differently!> by Kshitij Sobti

Sh erlock Holmes is often quoted as having said: “when you have eliminated the impossible, whatever remains, however improbable must be the truth.” It is this sentiment that most accurately conveys what constraint programming is all about.

It is not uncommon that the solution to a problem comes from a slow exploration of all possible answers, till we reach a tighter and tighter bound for what it could be, and finally arrive at a solution.

## Constraints lead to solutions?

Look at Sudoku, we're sure you are familiar with the popular number-based puzzle that adorns at least one corner of every newspaper in existence. If not, we'd love to buy the rock you've been living under, it seems like it might weather a nuclear explosion. If you seriously are not familiar with Sudoku, we'd advise you look it up before reading any further.

The general approach to Sudoku is to look for constraints. You'll tend to look at blank spots that are surrounded by many numbers, or have many numbers in the same row or column. You do this because those are the locations that are the most constrained and they have

the fewest possible options for numbers. You keep looking till you find a spot that can contain only one possible number, and you lock it down. With this new number in place, you suddenly have more constraints. You have new constraints for spaces in the same row, column or block. We now scan the puzzle again, looking for new spots that can now hold only one possible number. Harder Sudoku puzzles can take this a notch deeper by making you analyse a combination of constraints on multiple cells that give you a solution.

When you begin to think about it, there are many problems, and puzzles that work exactly the same way. Constraint programming is a concept that is designed to solve exactly such problems.

## So what exactly is constraint programming?

To have something to contrast constraint programming with, let's look at the immensely popular imperative programming. Imperative programming is the kind of programming commonly employed to create software written in C, Java, JavaScript, Python, Ruby etc. Most languages you might have encountered are imperative programming languages, or at least primarily support imperative

## \*coding matters



### \*io.js:

>>If you're a Node.js developer, get familiar with io.js. Where Node.js development seems to have stalled, io.js continues the project under a new name and a new open governance model.

<http://dgit.in/ioNodejs>



### \*Aurelia:

>>Welcome another JavaScript framework for building web applications for the world. Aurelia is a new modern, modular, self-contained web app framework that is built using latest (even unreleased) JavaScript features.

<http://dgit.in/NewAurelia>



### \*.Net open source

>>You've probably heard that Microsoft has made their .NET framework open source, and aim to support Linux and OSX for the same. You can now see the beginnings of this support by visiting their repositories.

<http://dgit.in/dotnotGit>